

Lab - Uso de escalado automático en AWS (Linux)

Dificultad:  Intermedio

Tiempo Estimado:  45 minutos

Servicios Principales:  Amazon EC2,  AWS CLI,  Elastic Load Balancing (ELB),  Amazon EC2 Auto Scaling,  Amazon CloudWatch.

Resumen y Objetivos

En este laboratorio usarás la Interfaz de Línea de Comandos de AWS (AWS CLI) para crear una instancia Amazon EC2 que aloje un servidor web y luego crearás una Amazon Machine Image (AMI) a partir de dicha instancia. Después, utilizarás esa AMI como base para desplegar un sistema que escale dinámicamente frente a picos de consumo apoyándote en Auto Scaling y distribuyendo el ingreso a las diferentes instancias en múltiples Zonas de Disponibilidad (AZs) albergadas bajo un Application Load Balancer (ALB).

Objetivos:

-  Crear una instancia EC2 utilizando AWS CLI.
 -  Crear una nueva AMI personalizada mediante programación con AWS CLI.
 -  Crear una plantilla de lanzamiento (Launch template) en Amazon EC2.
 -  Crear una configuración de lanzamiento de Auto Scaling.
 -  Configurar políticas de escalado y crear un grupo de Auto Scaling para mitigar de forma automatizada las cargas de tráfico variables.
-

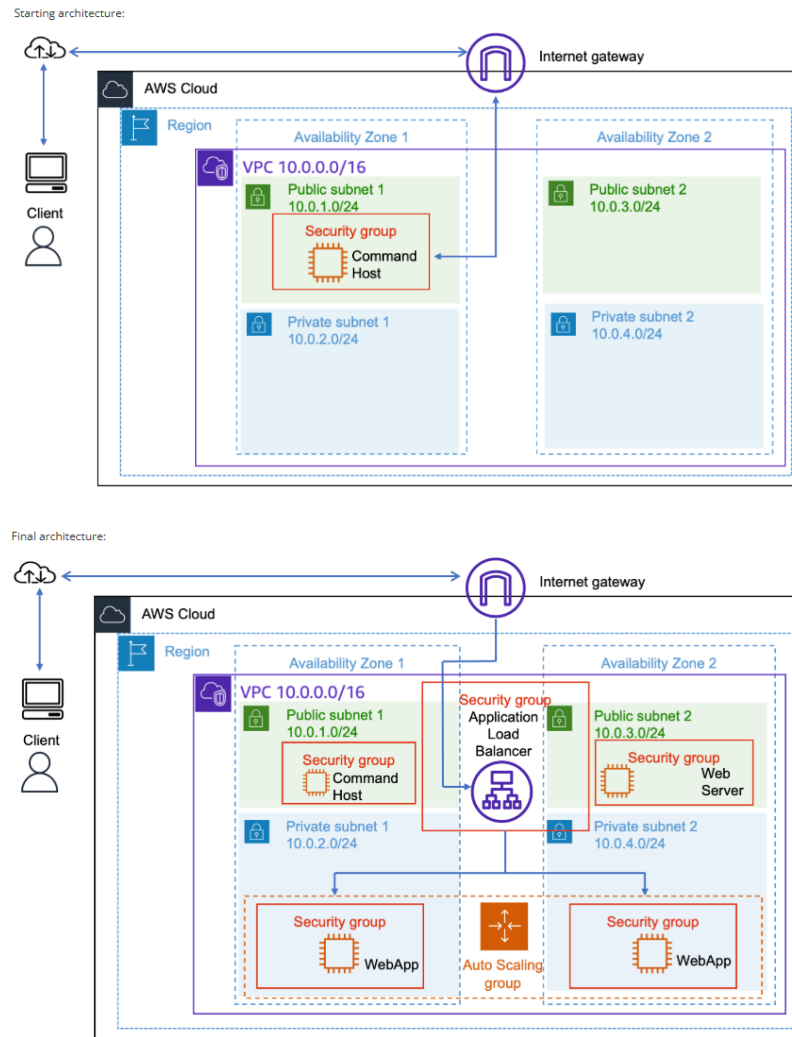
Análisis del Escenario

Se busca automatizar la preparación y aprovisionamiento de recursos. Ejecutar comandos a través de AWS CLI en lugar de la consola gráfica (UI) permite integrarlos velozmente a flujos de integración continua (CI/CD) usando de base una instancia de comandos autorizada a nivel IAM para actuar sobre AWS. Luego, se encapsula el servidor modelado en una AMI maestra para que Auto Scaling pueda proveer copias idénticas frente a la elasticidad requerida.

Todo este tráfico entrante público será segmentado por balanceador, permitiendo a los nodos ejecutarse seguros y ocultos en subredes privadas.

Arquitectura

-  **Arquitectura Inicial:** Una instancia de comando (*Command Host*) aprovisionada en una subred pública lista para que operes sobre la AWS CLI en un entorno seguro y delegues desde ahí.
-  **Arquitectura Final:** Application Load Balancer canalizando el tráfico de entrada de internet y repartiéndolo mediante un grupo de destino a múltiples instancias EC2 web, yacentes dentro de subredes privadas configuradas dinámicamente por un grupo de Auto Scaling en 2 Zonas de Disponibilidad distintas (Escalado In / Out apoyado en CloudWatch).



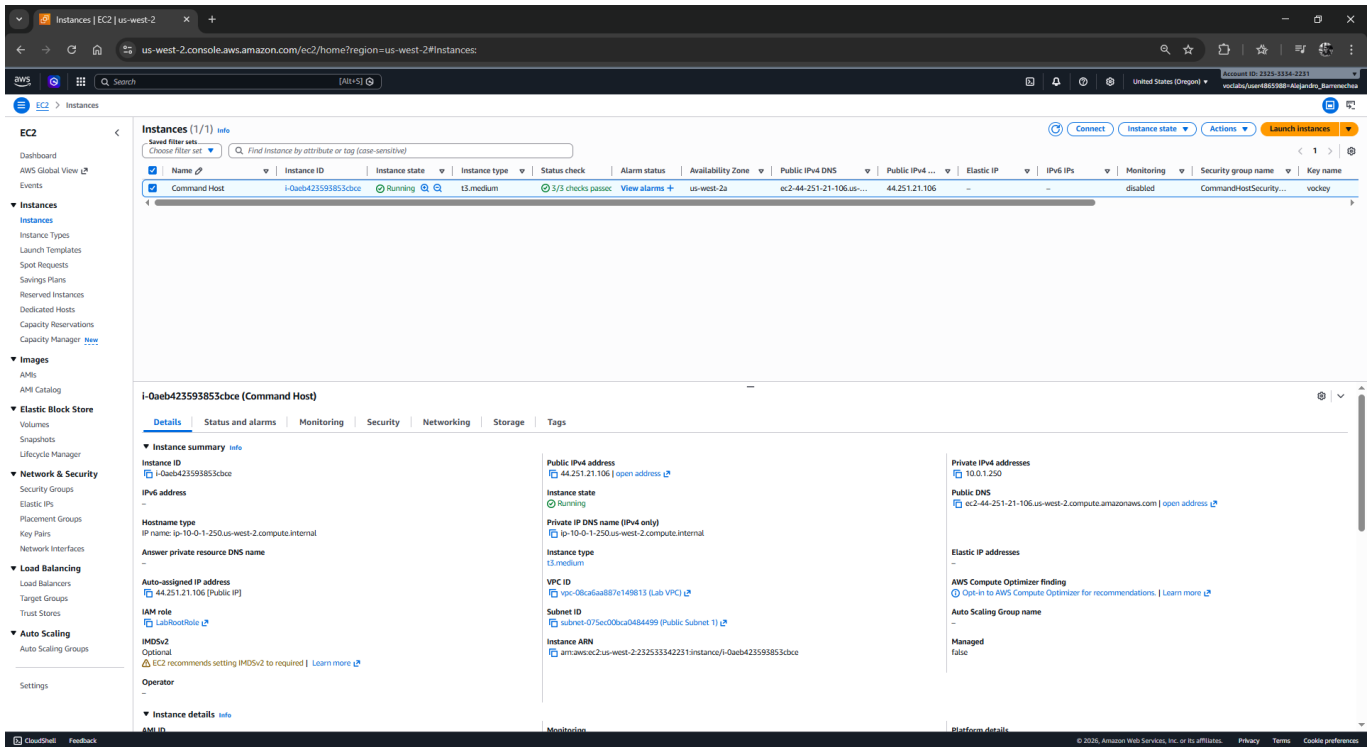
Desarrollo

Tarea 1: Creación de una AMI para el Auto Scaling mediante AWS CLI

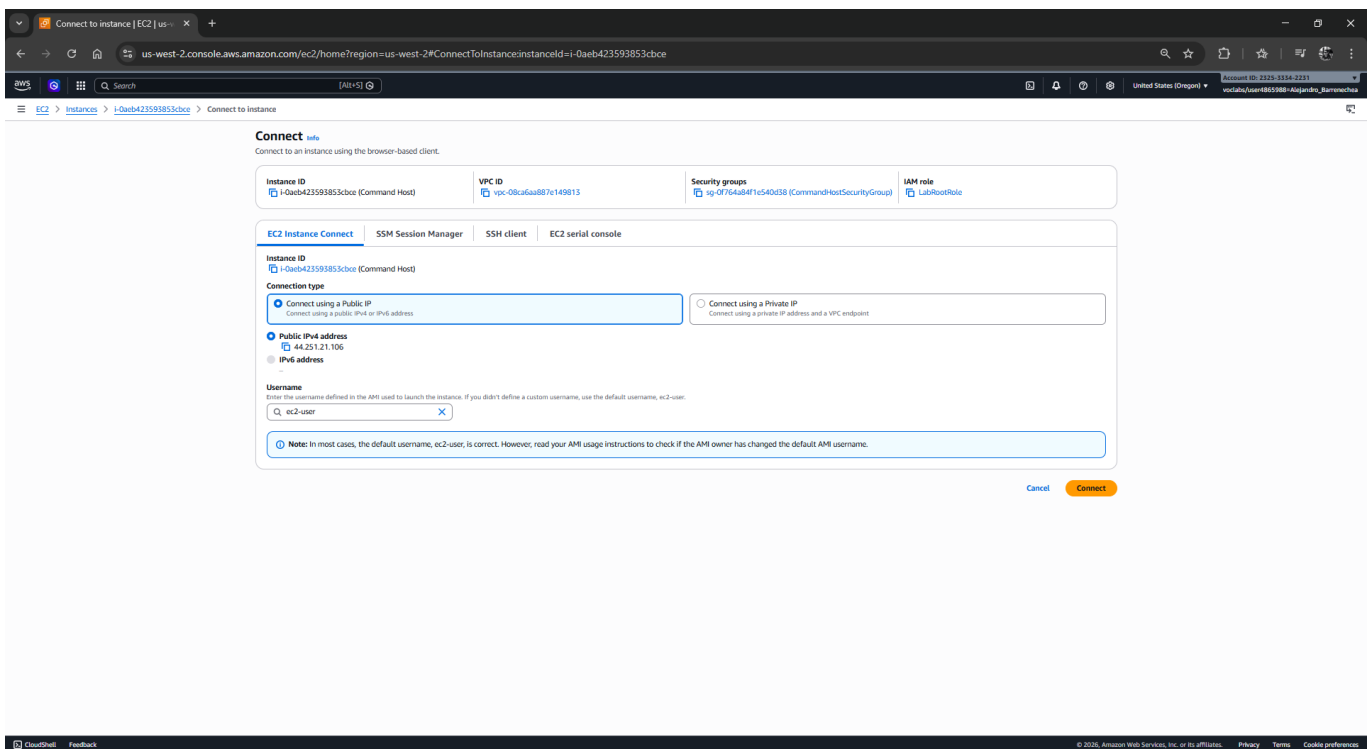
Usarás la máquina de comando (*Command Host*) para enviar instrucciones directas hacia AWS a través de la terminal, levantar un nuevo servidor con una app y generalizar su estado extrayendo una AMI.

Tarea 1.1: Conexión a la instancia Command Host

1. 🌐 En la consola de administración de AWS, dirígete a **EC2**.
2. 📁 En el menú lateral izquierdo entra en **Instancias**.
3. ☒ Selecciona la instancia **Command Host**.
4. 🖱️ Haz clic en el botón superior de **Conectar** (Connect).



5. Mantente predeterminado en la pestaña **EC2 Instance Connect** y presiona de nuevo en **Conectar**. Se te abrirá una terminal Linux en línea dentro del navegador.



Tarea 1.2: Configuración de la AWS CLI

La CLI ya está instalada, pero necesitamos actualizar sus datos del perfil para apuntar al mismo sector físico de tus recursos.

1. Para consultar explícitamente en qué Región estas trabajando, extrae los metadatos de AWS ejecutando:

```
curl http://169.254.169.254/latest/dynamic/instance-identity/document | grep
region
```

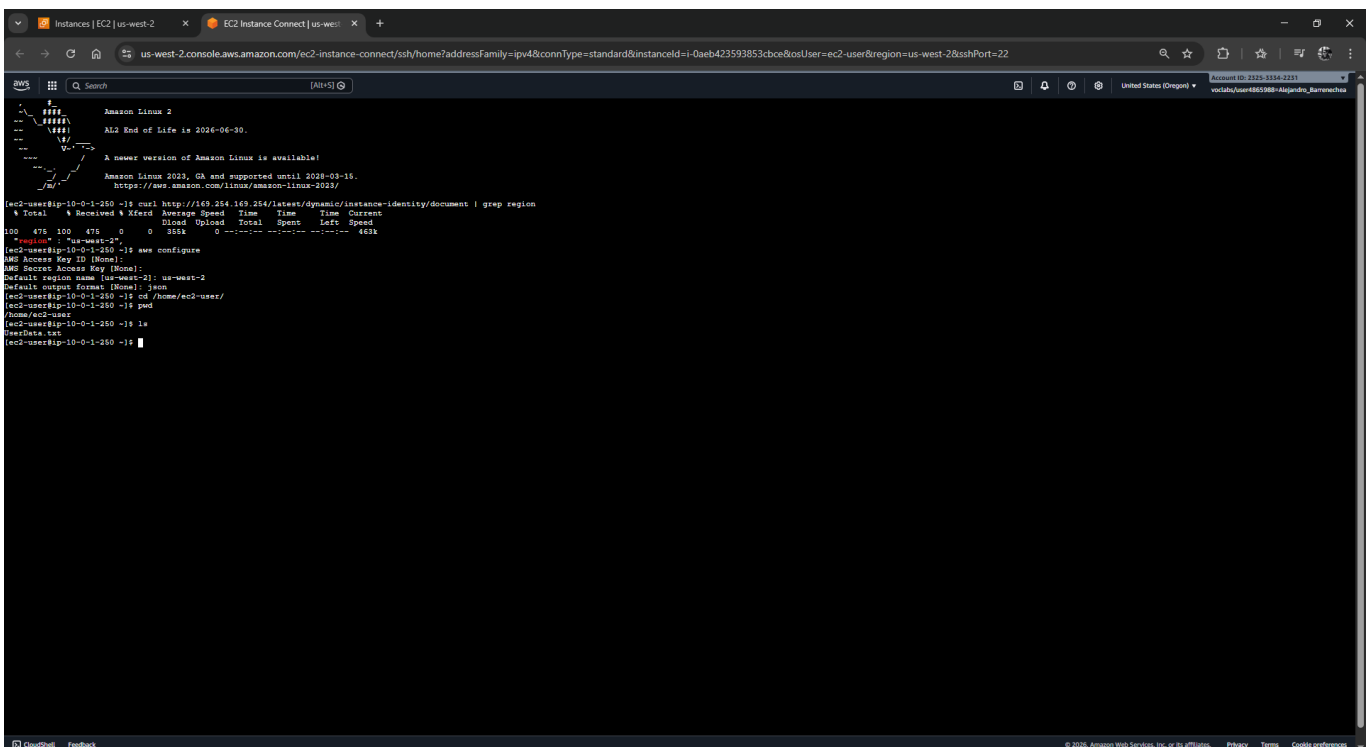
(Toma nota de ese valor de tu Región, por ejemplo: **us-west-2** o similar).

2.  Actualiza las credenciales de Amazon invocando la configuración:

```
aws configure
```

3.  Al ser consultado ingresa en el prompt:
 - **AWS Access Key ID:** (Presiona la tecla *Enter*, la CLI ya heredó el ID de Vocareum).
 - **AWS Secret Access Key:** (Presiona *Enter*).
 - **Default region name:** Ingresa la región que averiguaste previamente (ej. **us-west-2**) y presiona *Enter*.
 - **Default output format:** Escribe la palabra **json** y presiona *Enter*.
4.  Ubícate en tu propio directorio *Home* de scripts:

```
cd /home/ec2-user/
```



```

AWS
Search [Alt+S]
us-west-2.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-0aeb423593853bce&osUser=ec2-user&region=us-west-2&sshPort=22

Amazon Linux 2
---
  ____
--  /  __ \
-- /  /_  /
--/_____\
--
All End of Life is 2026-06-30.

A newer version of Amazon Linux is available!
Amazon Linux 2023, GA and supported until 2028-03-15.
https://aws.amazon.com/linux/amazon-linux-2023/

[ec2-user@ip-10-0-1-250 ~]$ curl http://169.254.169.254/latest/dynamic/instance-identity/document | grep region
  % Total    % Received % Xferd  Average Speed   Time    Time     Current
                                 Dload  Upload   Total   Spent    Left  Speed
100  475  100  475    0     0    475      0  0 --:--:-- --:--:-- --:--:-- 483k
[ec2-user@ip-10-0-1-250 ~]$ aws configure
AWS Access Key ID [None]:
AWS Secret Access Key [None]:
Default region name [us-west-2]: us-west-2
Default output format [None]: json
[ec2-user@ip-10-0-1-250 ~]$ cd /home/ec2-user/
[ec2-user@ip-10-0-1-250 ~]$ pwd
/home/ec2-user
[ec2-user@ip-10-0-1-250 ~]$ ls
awsData.txt
[ec2-user@ip-10-0-1-250 ~]$

```

Tarea 1.3: Creación de una nueva Instancia EC2 desde la CLI

1.  Explora el script de inicialización (**UserData.txt**) local a la máquina leyendo su contenido:

```
more UserData.txt
```

2.  A la izquierda externa a esa pestaña, verifica en tu entorno general del Lab, localiza la sección arriba **AWS details > Show**.
3.  Toma nota y copia a un bloc de texto local en tu computadora los valores exactos definidos en tu módulo para: **KEYNAME**, **AMIID**, **HTTPACCESS**, y **SUBNETID**. (Cierra el panel con X luego).
4.  Reemplaza minuciosamente las variables en MAYÚSCULAS en el siguiente comando con los valores que recién copiaste en el bloc de notas, cópialo ensamblado y arrójaló a tu terminal:

💡 **Nota analítica:** Observa lo ágil que es AWS CLI. Un solo comando asocia Llaves de conexión, define la máquina **t3.micro**, lanza los scripts al arrancar pre configurados, asocia una IP Pública, etiquetas (Tags) y devuelve el ID. Todo programáticamente automatizado.

```
EC2 Instance Connect | us-west-2 +  
us-west-2.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipV4&hostnameType=standard&instanceId=i-0ae6423593853cbe&ostUser=ec2-user&region=us-west-2&sshPort=22  
AWS Search [Alt+S]  
Amazon Linux 2  
--  
-- AI2 End of Life is 2026-04-30.  
--  
-- A newer version of Amazon Linux is available!  
--  
-- Amazon Linux 2023, GA and supported until 2028-03-15.  
-- https://aws.amazon.com/linux/amazon-linux-2023/  
  
[ec2-user@ip-10-0-1-250 ~]$ curl http://169.254.169.254/latest/dynamic/instance-identity/document | grep region  
& Total & Received & Xferd Average Speed Time Time Current  
Speed Upload Total Spent Left Speed  
100 % 495 100 % 0 0 355k 0 --:--:-- --:--:-- --:--:-- 463k  
  
#region "us-west-2"  
[ec2-user@ip-10-0-1-250 ~]$ aws configure  
AWS Access Key ID [None]:  
AWS Secret Access Key [None]:  
Default region name [us-west-2]: us-west-2  
Default output format [None]: json  
[ec2-user@ip-10-0-1-250 ~]$ cd /home/ec2-user/  
[ec2-user@ip-10-0-1-250 ~]$ pwd  
/home/ec2-user  
[ec2-user@ip-10-0-1-250 ~]$ ls  
UserData.txt  
[ec2-user@ip-10-0-1-250 ~]$ more UserData.txt  
#!/bin/bash  
  
yum update -y  
amazon-linux-extras install epel -y  
yum -y install httpd php stress  
systemctl enable httpd.service  
systemctl start httpd  
  
wget http://aws-cc-largeobjects.s3.amazonaws.com/CUR-TF-100-LAB-10-lab-autoscaling-linux/s3/ec2-stress.sh  
chmod ec2-stress.sh  
  
echo "UserData has been successfully executed." >> /home/ec2-user/result  
find -wholename /root/.history -wholename /home/*/.history exec rm -f {} ;  
find / -name "authorized_keys" -exec rm -f {} ;  
rm -rf /var/lib/cloud/data/scripts/*  
[ec2-user@ip-10-0-1-250 ~]$ aws ec2 run-instances --key-name voyceky --instance-type t3.micro --image-id ami-0a7c9eef70b254dc --user-data file:///home/ec2-user/UserData.txt --security-group id sg-05e97a420c20782d5 --subnet id subnet-01245e45070b2dd5 --associate-public-ip-address  
t-02E0a3194d4489c  
[ec2-user@ip-10-0-1-250 ~]$ CC  
[ec2-user@ip-10-0-1-250 ~]$
```

```
aws ec2 wait instance-running --instance-ids NEW-INSTANCE-ID
```




🕒 Tarea 1.4: Crear una AMI Base Personalizada (Custom AMI)

Basa tu futuro autoscale en el esqueleto actual inmutable para clonaje.

1. 🖥 De regreso a tu terminal de Linux EC2, ejecuta empaquetar una Amazon Machine Image referenciando el **NEW-INSTANCE-ID**:

```
aws ec2 create-image --name WebServerAMI --instance-id NEW-INSTANCE-ID
```

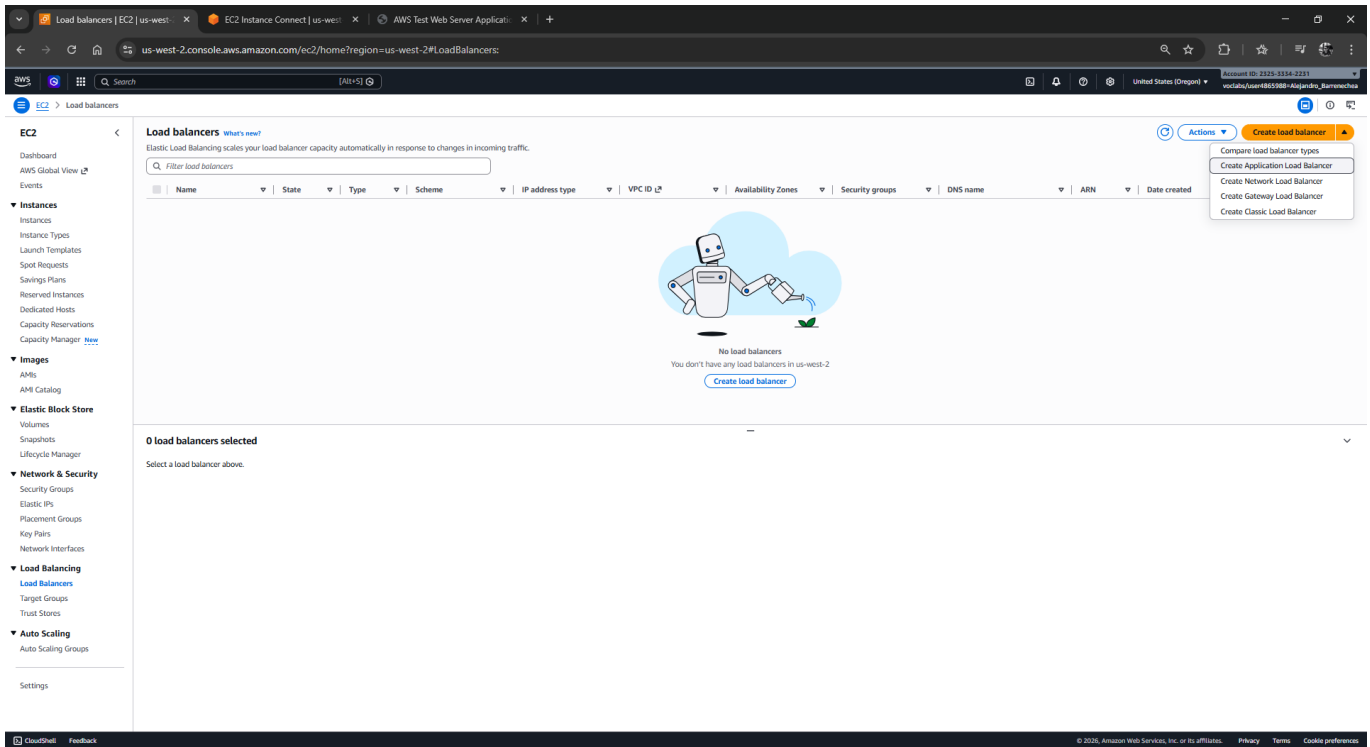
*(Opcionalmente esto reinicia ligeramente la máquina madre apuntada del **instance-id** vaciando memorias volátiles pre-instantánea a nivel de hypervisor, protegiendo así la integridad de archivos disco duro base).*

🌐 Tarea 2: Entorno de escalado automático desde la Consola gráfica (AWS Management Console)

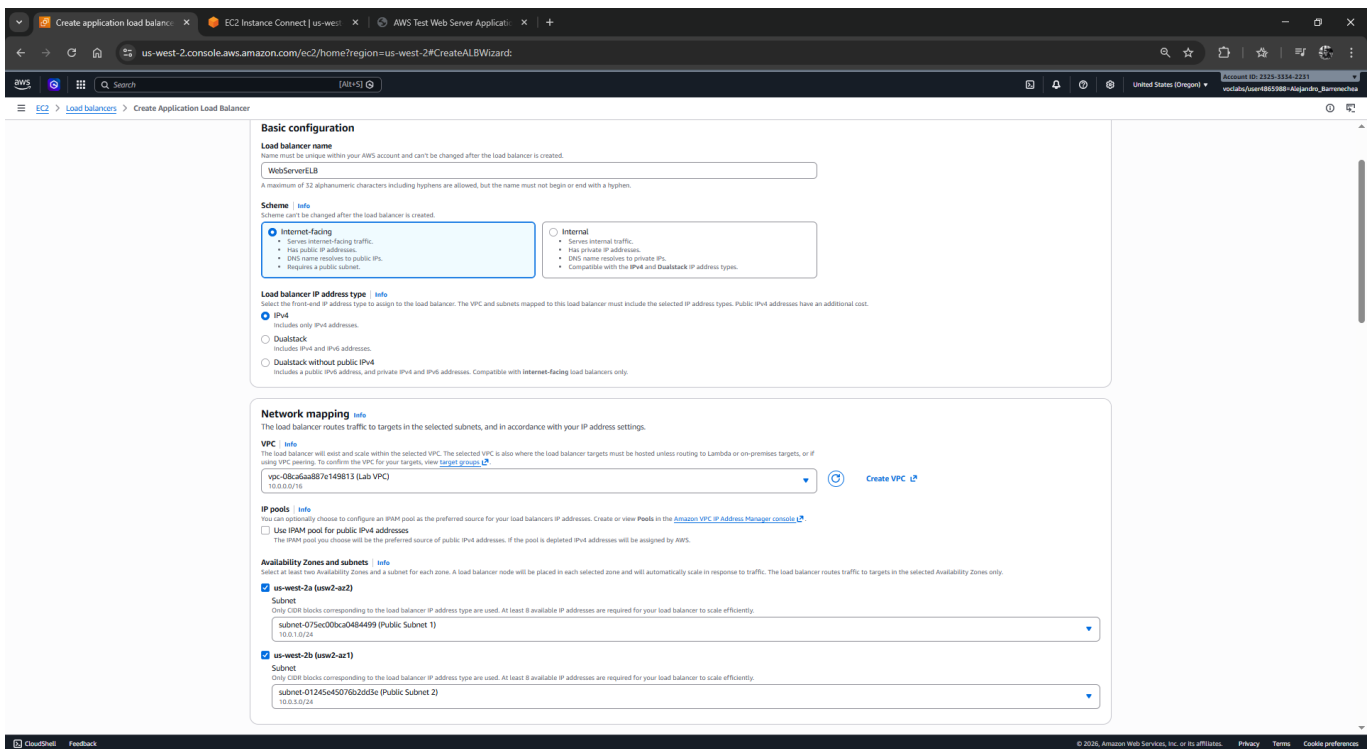
Mientras tu AMI se destila en los discos rígidos en background de la nube, pasarás a gestionar la infraestructura desde el entorno visual de la consola AWS.

🔧 Tarea 2.1: Crear un Application Load Balancer

1. 🕒 Desde la consola general EC2 en el navegador, ve abajo en la columna de la izquierda hacia **Balanceadores de carga**.
2. + Selecciona el botón **Crear balanceador de carga**.
3. 📄 Elige la tarjeta **Application Load Balancer** y oprime **Crear**.



4. ⚙️ Para **Nombre del balanceador de carga**, nómbralo **WebServerELB**.
5. 🏠 En **Mapeo de Red**, elige del menú desplegable **Lab VPC**.
6. 📍 Marca las 2 zonas de disponibilidad que te asoman y escoge la cara de red pública:
 - Para la primera, elije **Public Subnet 1**.
 - Para la segunda, elija la **Public Subnet 2**.



7. 🛡️ En **Grupos de seguridad**: marca la X para remover el predeterminado. Elige el ya prehecho e identificado como **HTTPAccess**.
8. 🎧 En **Agentes de escucha y enrutamiento**: Haz clic en el texto o enlace **Crear grupo de destino** (abrirá una pestaña nueva).

Security groups [info](#)
A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups
Select up to 5 security groups
[HTTPAccess \(sg-05d97a42b20782d5\)](#)

Listeners and routing [info](#)
A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

Listener HTTP:80 [Remove](#)

Protocol
HTTP

Port
80
1-65535

Default action [info](#)
The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action
☒ Forward to target groups ☐ Redirect to URL ☐ Return fixed response

Forward to target group [info](#)
Choose a target group and specify routing weight or [create target group](#).

Target group
Select a target group

Weight
1
0-999

Percent
100%

+ Add target group
You can add up to 4 more target groups.

Target group stickiness [info](#)
Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.

☐ Turn on target group stickiness

Listener tags - optional
Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

[Add listener tag](#)
You can add up to 50 more tags.

[Add listener](#)
You can add up to 49 more listeners.

9. Dentro de tu flamante pestaña Target Groups para *Configuración Básica*:

- **Tipo de destino:** Selecciona explícitamente **Instancias**.
- **Nombre de grupo de destino:** **webserver-app**

Create target group
A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - immutable
Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.

Target type
Indicate what resource type you want to target. Only this selected resource type can be registered to this target group.

☒ **Instances**
Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or EC2 instances for automatic management.
Suitable for: **ALB** **NLB** **GWLB**

☐ **IP addresses**
Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
Suitable for: **ALB** **NLB** **GWLB**

☐ **Lambda function**
Supports load balancing to a single Lambda function. ALB required as traffic source.
Suitable for: **ALB**

☐ **Application Load Balancer**
Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
Suitable for: **NLB**

Target group name
Name must be unique per Region per AWS account.
webserver-app
Accept: a-z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 13/32

Protocol
Protocol for communication between the load balancer and targets.
HTTP

Port
Port number where targets receive traffic. Can be overridden for individual targets during registration.
80
1-65535

IP address type
Only targets with the indicated IP address type can be registered to this target group.

☒ **IPv4**
Each instance has a default network interface (eni) that is assigned the primary private (IPv4) address. The instance's primary private (IPv4) address is the one that will be applied to the target.

☐ **IPv6**
Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eni). [Learn more](#)

VPC
Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.
vpc-08ca6a8b7e149813 (Lab VPC)
10.0.0.0/16

[Create VPC](#)

Protocol version
☒ **HTTP/1**
Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

☐ **HTTP/2**
Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.

☐ **gRPC**
Send requests to targets using gRPC. Supported when the request protocol is gRPC.

- **Inspecciones de estado** (*Health check path*): Introduce manualmente en la ruta o Path **/index.php** (crucial para apuntar a la base de la app correcta).
- Ve al fondo y dale **Siguiente** (Next).

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-08ca6aa887e149813 (Lab VPC)
10.0.0.0/16

[Create VPC](#)

Protocol version

☒ HTTP1
Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

☐ HTTP2
Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.

☐ gRPC
Send requests to targets using gRPC. Supported when the request protocol is gRPC.

Health checks
The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.

Health check protocol
HTTP

Health check path
Use the default path of "/" to perform health checks on the root, or specify a custom path if preferred.
/index.php
Up to 1024 characters allowed.

Advanced health check settings

Target optimizer - optional
Use a target control port when the target has a strict concurrency limit.

Target control port
The port on which the target communicates its capacity. This value can't be modified after target group creation.
Enter target control port
Valid range: 1-65535

Attributes
Certain default attributes will be applied to your target group. You can view and edit them after creating the target group.

Tags - optional
Consider adding tags to your target group. Tags enable you to categorize your AWS resources so you can more easily manage them.

[Cancel](#) [Next](#)

10. En la página de *Registrar destinos* (Register Targets), no hagas *nada* manual. Haz clic en **Siguiente** (Auto Scaling se encargará de las instancias).

Register targets - recommended
This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (2)

Instance ID	Name	State	Security groups	Zone	Private IPv4 address	Subnet ID	Launch time
i-025f0a3134c14488c	WebServer	Running	HTTPAccess	us-west-2b	10.0.3.72	subnet-01245e45076b2d35e	April 3, 2026, 18:34 (UTC-03:00)
i-0aeb423593853c3ce	Command Host	Running	CommandHostSecurityGroup	us-west-2a	10.0.1.250	subnet-075ec08ca0484499	April 3, 2026, 18:08 (UTC-03:00)

0 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.
80
Y-65535 (separate multiple ports with comma)
[Include as pending below](#)

Review targets

Targets (0)
Filter targets
Show only pending
Remove all pending

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet ID	Launch time
No instances added yet Specify instances above, or leave the group empty if you prefer to add targets later.								

0 pending

[Cancel](#) [Previous](#) [Next](#)

11. En la página conclusiva *Revisar y crear* haz clic final en **Crear grupo de destino**.

Review and create
Review your target group configuration before creating

Step 1: Target group details [Edit](#)

Name webserv-app	Target type Instance	Protocol : Port HTTP: 80	Protocol version HTTP1
VPC vpc-08ca6a887e149813 ↗	IP address type IPv4		

Health check details

Health check protocol HTTP	Health check path /index.php	Health check port traffic-port	Interval 30 seconds
Timeout 5 seconds	Healthy threshold 5	Unhealthy threshold 2	Success codes 200

Step 2: Register targets [Edit](#)

Targets (0)

Instance ID	Name	Port	Zone
No targets added			

[Cancel](#) [Previous](#) [Create target group](#)

12. Cierra esta recién creada pestaña y en la primaria e inconclusa ventana del **Load Balancer**, usa el botón circular de Actualizar () y localiza para anexar el nuevo recurso listado: **webserv-app**.
13. Termina todo el proceso al final de página presionando **Crear balanceador de carga**.
14. Oprime posteriormente en **Ver balanceador de carga** para seleccionar tu ALB y asegurarte de copiar su **Nombre DNS** que oficiará de entrada oficial a la internet.

Tarea 2.2: Creación de la Plantilla de lanzamiento

1. Navega en el margen izquierdo bajo *Instancias* a **Plantillas de lanzamiento** (Launch Templates).
2. Haz clic en el banner naranja **Crear plantilla de lanzamiento**.

EC2 launch templates
Streamline, simplify and standardize instance launches

Use launch templates to automate instance launches, simplify permission policies, and enforce best practices across your organization. Save launch parameters in a template that can be used for on-demand launches and with managed services, including EC2 Auto Scaling and EC2 Fleet. Easily update your launch parameters by creating a new launch template version.

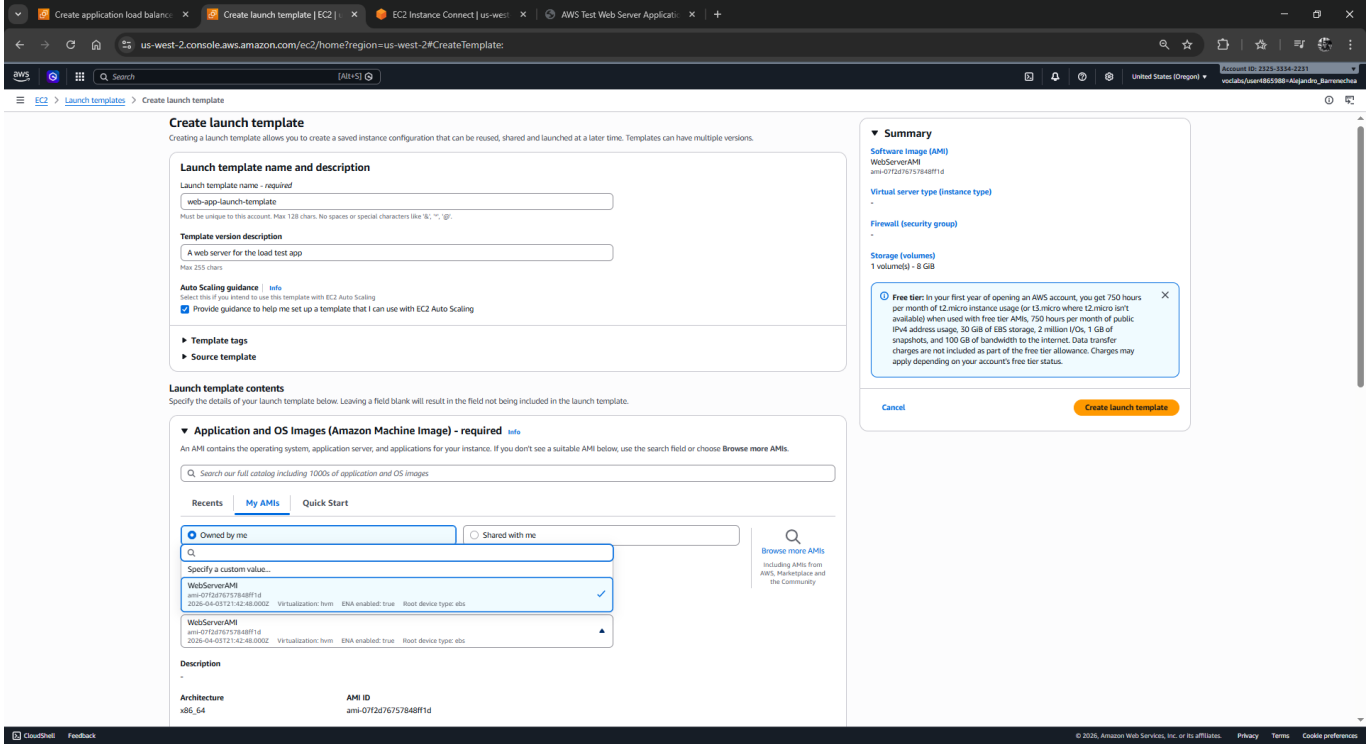
Benefits and features

- Streamline provisioning**
Minimize steps to provision instances. With EC2 Auto Scaling, updates to a launch template can be automatically passed to an Auto Scaling group. [Learn more](#)
- Simplify permissions**
Create shorter, easier to manage IAM policies. [Learn more](#)
- Governance**
Ensure best practices are used across your organization. [Learn more](#)

New launch template
[Create launch template](#)

Documentation
[Documentation](#)
[API reference](#)

3.  Configura **Nombre de plantilla de lanzamiento**: **web-app-launch-template** y **Descripción**: **A web server for the load test app.**
 - Activa el *checkbox* de **Proporcionar orientación para ayudarme a configurar...** bajo *Autoscaling guidance.*
4.  En *Imágenes de aplicación y SO*, pestaña **Mis AMI**: Selecciona tu AMI generada en consola de comandos anteriormente: **WebServerAMI**.



Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - required
web-app-launch-template

Must be unique to this account. Max 128 chars. No spaces or special characters like '<', '>', '@', etc.

Template version description
A web server for the load test app

Max 255 chars

Auto Scaling guidance | info
Select this if you intend to use this template with EC2 Auto Scaling.
☒ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

Template tags
Source template

Launch template contents

Specify the details of your launch template below. Leaving a field blank will result in the field not being included in the launch template.

Application and OS Images (Amazon Machine Image) - required | info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents | **My AMIs** | Quick Start

Owned by me | Shared with me

Specify a custom value...

WebServerAMI
ami-072d76757848f1d
2026-04-03T21:42:48.000Z | Virtualization: hvm | EBS enabled: true | Root device type: ebs

WebServerAMI
ami-072d76757848f1d
2026-04-03T21:42:48.000Z | Virtualization: hvm | EBS enabled: true | Root device type: ebs

Description

Architecture | x86_64 | AMI ID | ami-072d76757848f1d

Summary

Software Image (AMI)
WebServerAMI
ami-072d76757848f1d

Virtual server type (instance type)
-

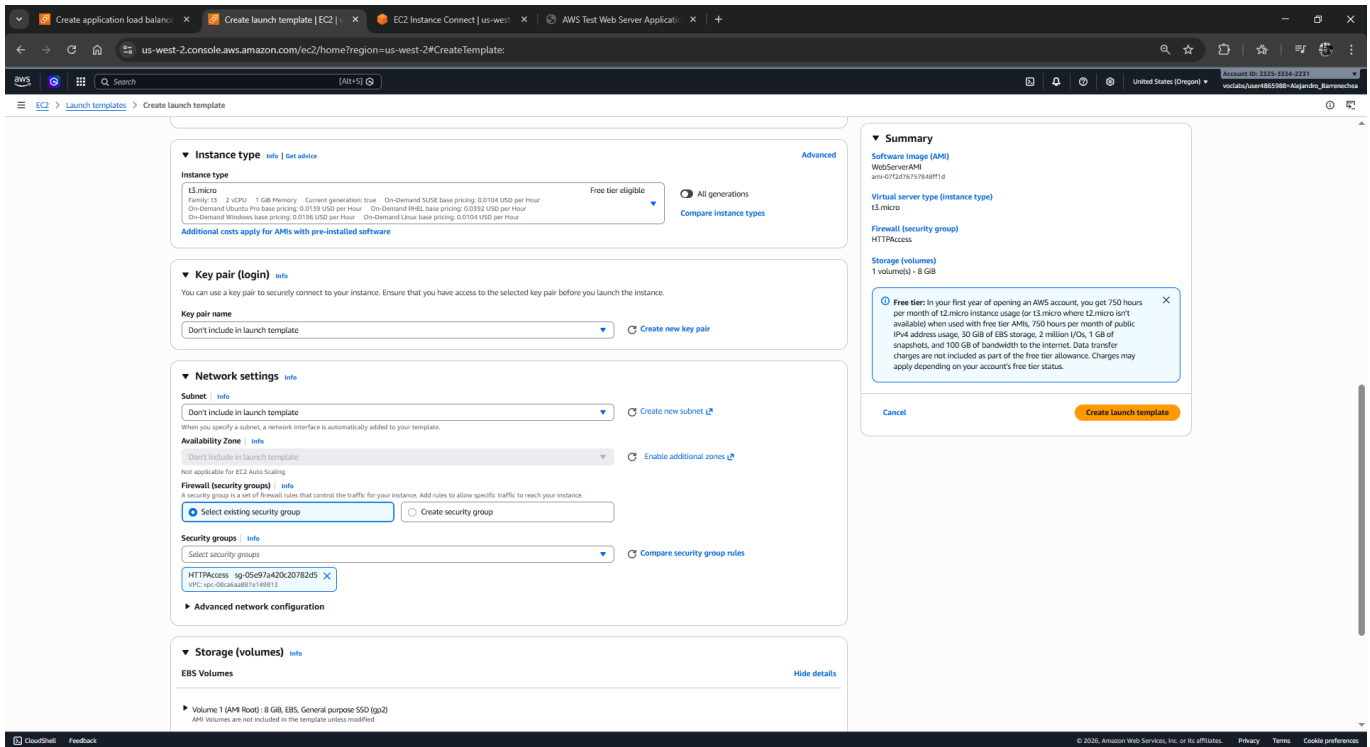
Firewall (security group)
-

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GiB of snapshots, and 100 GiB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance. Charges may apply depending on your account's free tier status.

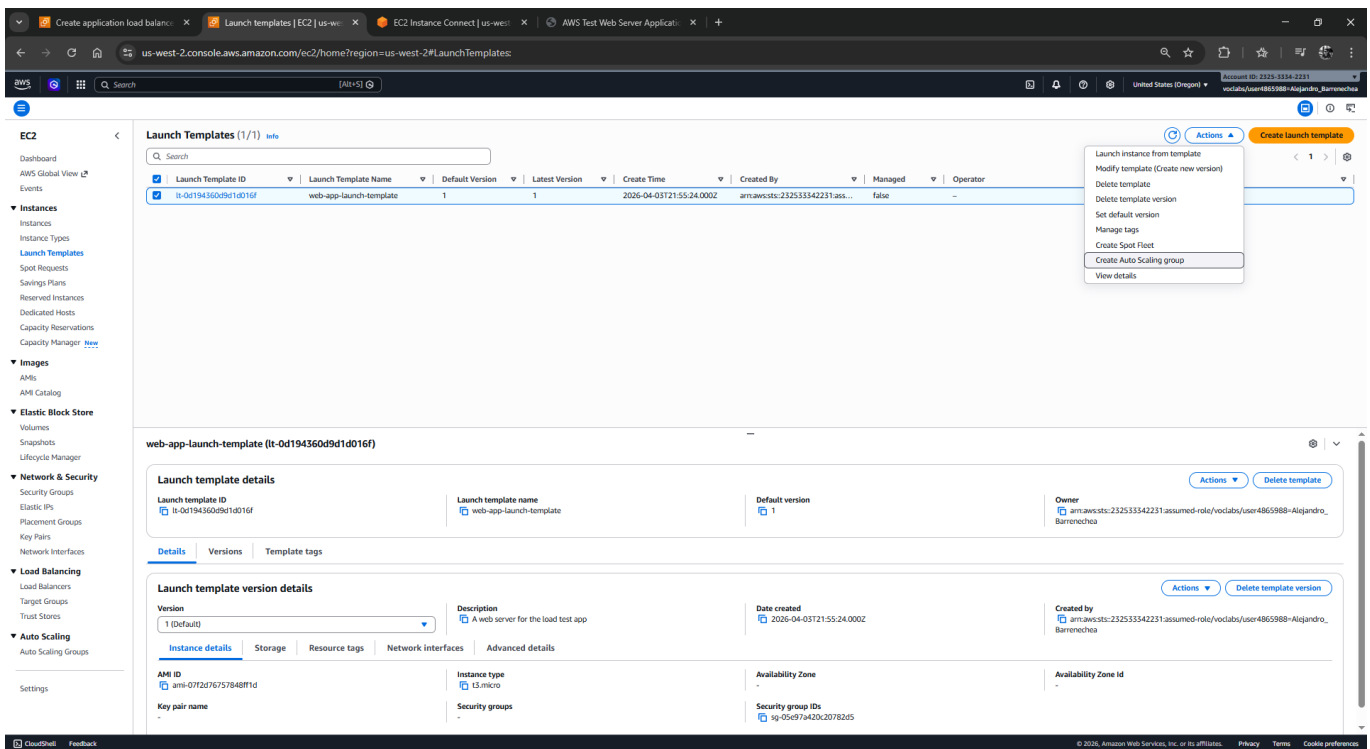
Cancel | **Create Launch template**

5.  En **Tipo de instancia**: Busca por menú y anexa la familia **t3.micro**.
6.  En **Par de claves (inicio de sesión)**: Determinálo estrictamente marcando *No incluir en la plantilla de lanzamiento*.
7.  En **Configuraciones de red > Grupos de Seguridad**: Selecciona la opción idónea **HTTPAccess**.
8.  Al fondo de todo oprime **Crear plantilla de lanzamiento**. (Confirma mirando un éxito indicando **Successfully created...**).

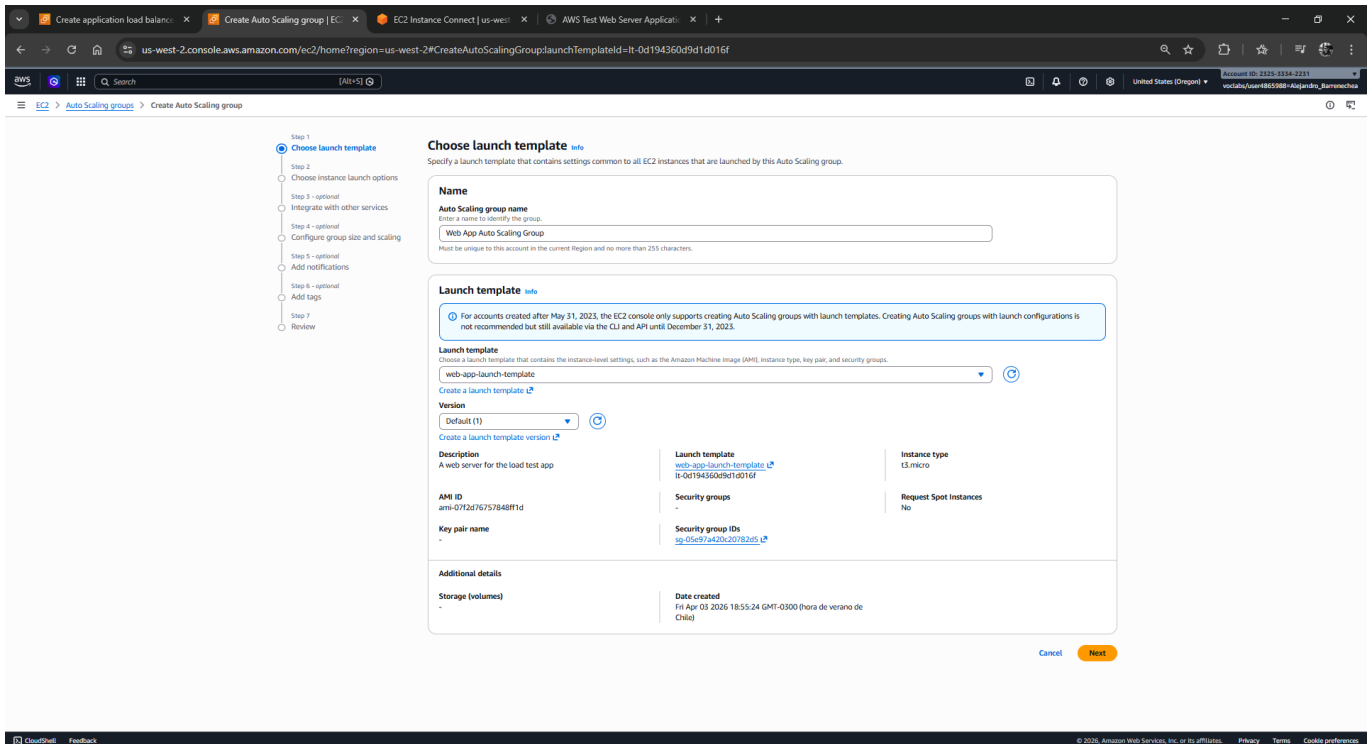


☑ Tarea 2.3: Configurar el Grupo de Auto Scaling

- ☑ Haz clic en "Ver plantillas de lanzamiento". Tilda tu nueva **web-app-launch-template** a modo selección simple. Accede en la zona superior de **Acciones**, elige del menú **Crear Grupo de Auto Scaling**.

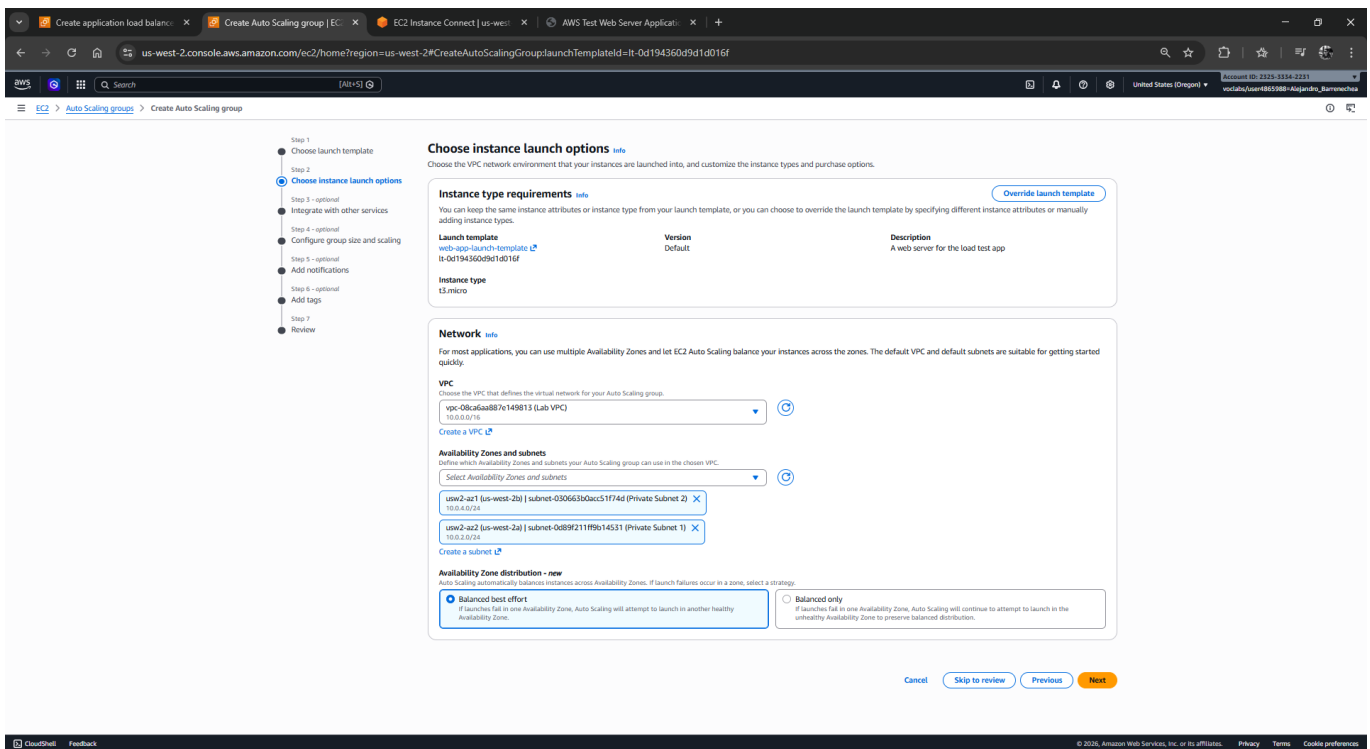


- ✎ Asigna el valor del nombre del contenedor: **Web App Auto Scaling Group** (en Nombre de grupo...). Usa **Siguiente**.



3. Para las selecciones de RED:

- Apunta a tu **VPC**: Lab VPC.
- Zonas de subredes: selecciona únicamente **Private Subnet 1 (10.0.2.0/24)** y **Private Subnet 2 (10.0.4.0/24)**. Esto asegura abstracción total frente a la nube abierta escondiendo nuestros nodos de bases de datos o computo. Haz clic en **Siguiente**.



4. Funciones Avanzadas y de Balanceador:

- En Balanceo de Carga selecciona: **Asociar a un balanceador de carga existente**.
- Haz clic en **Elegir de los grupos de destino...** y selecciona el link a tu Target de grupo **webserver-app | HTTP**.

- Marca la casilla *Activar comprobación de estado de ELB* de igual forma en la inspección médica. Acepta y ve a **Siguiente**.

Load balancing info
Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

Select Load balancing options

☐ No load balancer
Traffic to your Auto Scaling group will not be fronted by a load balancer.

☒ Attach to an existing load balancer
Choose from your existing load balancers.

☐ Attach to a new load balancer
Quickly create a basic load balancer to attach to your Auto Scaling group.

Attach to an existing load balancer

Select the load balancers to attach
This option allows you to attach Application, Network, or Gateway Load Balancers.

☒ Choose from your load balancer target groups
Choose from your existing load balancers.

☐ Choose from Classic Load Balancers

Existing load balancer target groups
Only instance target groups that belong to the same VPC as your Auto Scaling group are available for selection.

Select target groups

webserver-app | HTTP
Load balancer: Not associated with any load balancer

One of your target groups is not yet associated with any load balancer. In order for routing and scaling to occur, you will need to attach the target group to a load balancer. This can be done later in the [Load Balancing console](#).

VPC Lattice integration options info
To improve networking capabilities and scalability, integrate your Auto Scaling group with VPC Lattice. VPC Lattice facilitates communications between AWS services and helps you connect and manage your applications across compute services in AWS.

Select VPC Lattice service to attach

☒ No VPC Lattice service
VPC Lattice will not manage your Auto Scaling group's network access and connectivity with other services.

☐ Attach to VPC Lattice service
Incoming requests associated with specified VPC Lattice target groups will be routed to your Auto Scaling group.

[Create new VPC Lattice service](#)

Application Recovery Controller (ARC) zonal shift new info
During an Availability Zone impairment, target instance launches towards other healthy Availability Zones.

☐ Enable zonal shift
New instance launches will be reintegrated towards healthy Availability Zones until the zonal shift is cancelled.

Health checks
Health checks increase availability by replacing unhealthy instances. When you use multiple health checks, all are evaluated, and if at least one fails, instance replacement occurs.

EC2 health checks
[Always enabled](#)

Additional health check types - optional info
☒ Turn on Elastic Load Balancing health checks Recommended

- Configuración de las métricas de variabilidad del grupo:
 - Capacidad Deseada (*Desired*): 2
 - Capacidad Mínima (*Min*): 2
 - Capacidad Máxima (*Max*): 4
- Más abajo, en *Políticas de escalado*, selecciona a la orden de **Política de escalado de seguimiento de destino** (*Target Tracking scaling policy*).
 - Constata que siga en Uso Promedio de CPU, y setea el margen del **Valor de destino al 50** (50%). Pulsa **Siguiente**.

Configure group size and scaling - optional info
Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size info
Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type
Choose the unit of measurement for the desired capacity value. vCPUs and MemoryGB are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances)

Desired capacity
Specify your group size.

2

Scaling info
You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits
Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity **Max desired capacity**

2 4

Equal or less than desired capacity Equal or greater than desired capacity

Automatic scaling - optional
Choose whether to use a target tracking policy info
You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☐ No scaling policies
Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☒ Target tracking scaling policy
Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Scaling policy name
Target Tracking Policy

Metric type info
Monitored metric that determines if resource utilization is too low or high. If using EC2 metrics, consider enabling detailed monitoring for better scaling performance.

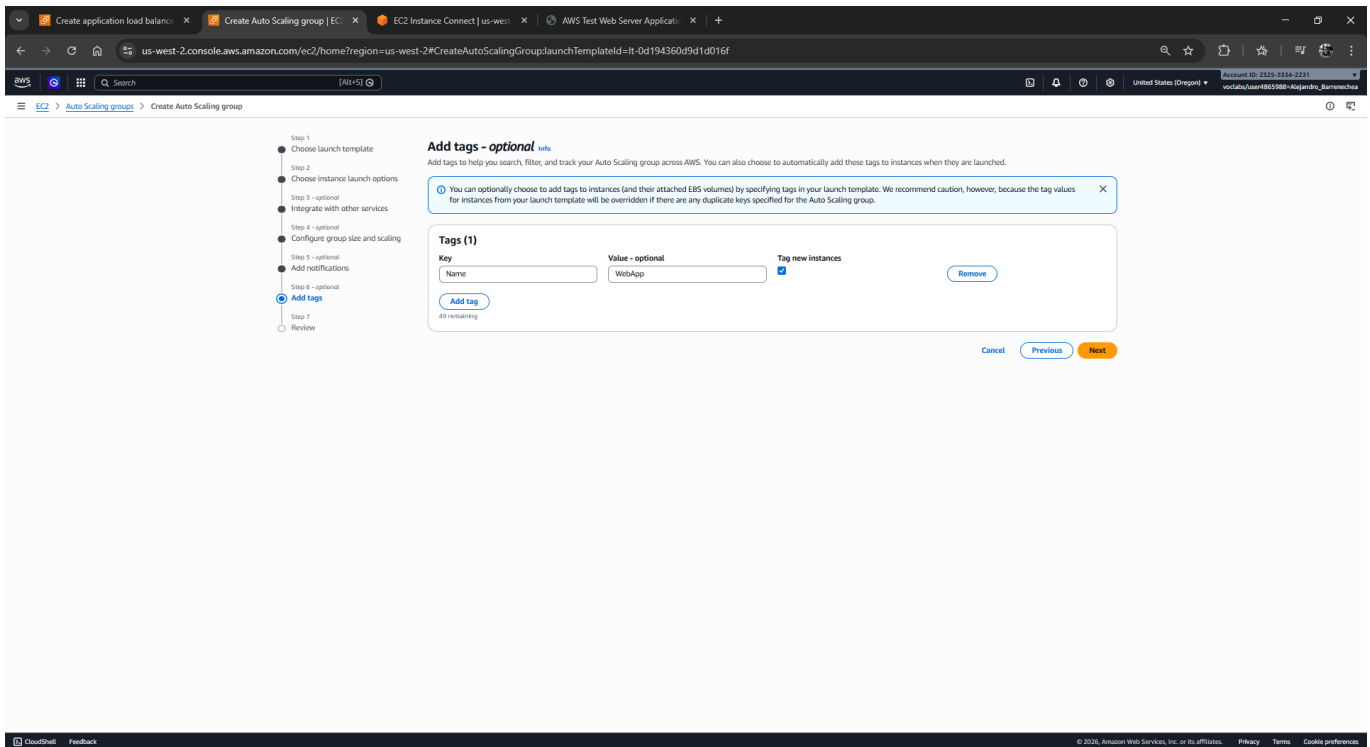
Average CPU utilization

Target value
50

Instance warmup info
300 seconds

☐ Disable scale in to create only a scale-out policy

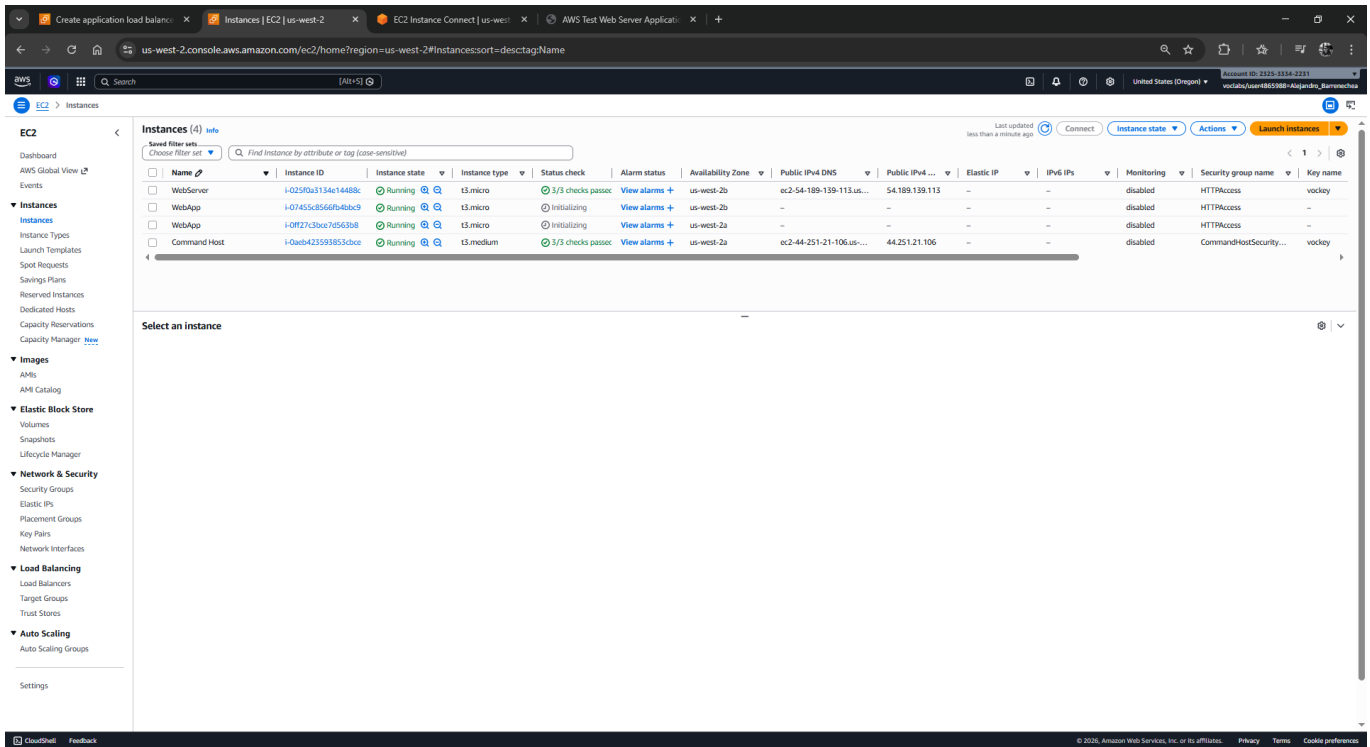
7.  Salta Notificaciones con **Siguiente**.
8.  Inserta metadatos útiles en las etiquetas clicando "Agregar etiqueta": Clave (Key): **Name**. Valor (Value): **WebApp**. **Siguiente**.



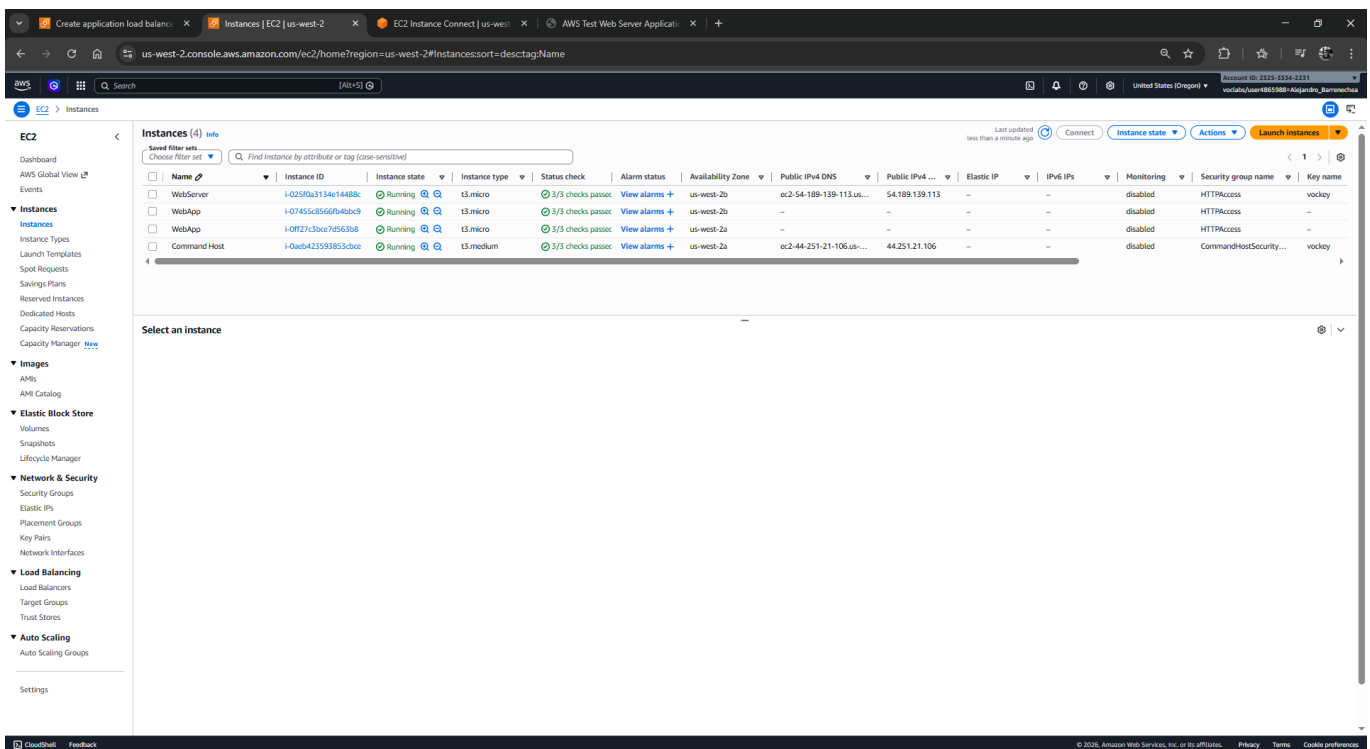
9.  Finaliza presionando **Crear grupo de Auto Scaling**. (Tu grupo ha sido ordenado, las dos copias iniciales de instancias están preparadas para inyectarse de inmediato por detrás dentro de entornos de subred privada).

Tarea 3: Verificación de Estado en conjunto Autoscaling / Load Balancer

1.  Ve al panel maestro izquierdo y explora la visibilidad en **Instancias**. Deberás presenciar dos máquinas "WebApp" arrancando. Como notarás el *Status Check* dirá algo como "Inicializando...".



2. ⌛ No las pierdas de vista (usando Refresh / Actualizar 🔄). Tu indicador verde es cuando las dos lleguen pacientemente en su chequeo a indicar **2/2 Checks passed**.



3. 🐞 El Balanceador debe poder sondear esas instancias ahora. Ve a tu panel bajo *Balanceo de Carga* y haz clic en **Grupos de Destino** (Target Groups), tocando arriba de tú **webserver-app**.
4. 🌐 Posiciónate en la pestaña superior de **Destinos** (Targets); si Auto Scaling fue exitoso, el balanceador ya los conoce y muestra listados. Al principio tendrán Health status como *Initial*, luego si recargas por unos minutos pasarán todas finalmente a arrojar test exitoso y saludable: **Healthy**.

The screenshot shows the AWS Management Console interface for the 'Target groups' page. The left sidebar contains navigation links for various AWS services. The main content area shows the 'Target groups' page for the 'webserver-app' target group. The 'Targets' tab is selected, displaying a table of registered targets. The table has columns for Instance ID, Name, Port, Zone, Health status, and Health status details. Two targets are listed, both with a health status of 'Initial' and 'Target registration is in progress'.

Instance ID	Name	Port	Zone	Health status	Health status details	Admin...	Overrid...	Launch...	Anomaly detection result
i-07455c8566fb4b3c9	WebApp	80	us-west-2b (usw2-az1)	Initial	Target registration is in progress	No override...	No overrid...	April 3, 20...	Normal
i-0f2773bce7d563b8	WebApp	80	us-west-2a (usw2-az2)	Initial	Target registration is in progress	No override...	No overrid...	April 3, 20...	Normal

Tarea 4: Someter la arquitectura probando una Escalada Dinámica a demanda

Como validación, deberás emular un comportamiento que amenace colapsar un nodo de computo provocando la reacción programada previamente de CPU y AWS CloudWatch.

1. 🌐 Abre tu navegador en pestaña externa. Copia e ingresa la **URL DNS de tu balanceador de carga** (ELB) guardada anteriormente, y accede con un Enter.
2. ⚡ En el sitio visualizado oprime velozmente el mandato o botón en la interfaz web de **Start Stress**.
(Por detrás, una solicitud a base de loops infinitos está corriendo para que tu instancia empuje la CPU a reportar cerca del 100% al sistema).

The screenshot shows a web browser window with the URL 'ec2-54-189-139-113.us-west-2.compute.amazonaws.com/index.php?stress=start'. The page displays the 'Start Stress' button, which is used to generate load on the application.

Generating load



- Regresa al Dashboard de la plataforma Amazon EC2. Haz Scroll abajo sobre el panel al segmento *Auto Scaling* hacia **Grupos de Auto Scaling** y selecciona tu **Web App Auto Scaling Group**.
- En el desglose detallado de ese grupo asila la pestaña vinculada a **Actividad** (Activity).

Web App Auto Scaling Group

Web App Auto Scaling Group Capacity overview

Desired capacity	Scaling limits	Desired capacity type	Status
2	2 - 4	Units (number of instances)	-

Date created: Fri Apr 03 2026 19:03:03 GMT-0300 (hora de verano de Chile)

Activity history (2)

Status	Description	Cause	Start time	End time
Successful	Launching a new EC2 instance i-07455c8566fb4bcb9	At 2026-04-03T22:05:03Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 2. At 2026-04-03T22:05:08Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 2.	2026 April 03, 07:03:10 PM -03:00	2026 April 03, 07:05:15 PM -03:00
Successful	Launching a new EC2 instance i-07455c8566fb4bcb9	At 2026-04-03T22:05:03Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 2. At 2026-04-03T22:05:08Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 2.	2026 April 03, 07:03:10 PM -03:00	2026 April 03, 07:05:15 PM -03:00

- Al igual que las alarmas de Cloudwatch, el Auto Scaling tardará aproximadamente de tres a cinco plenos minutos. Puedes apretar actualización (Refresh) o ver en paralelo el menú de *Cloudwatch - Alarmas*. (Notarás como producto de rebasar la marca estadística del 50%, AWS lanza de manera automática sin intervención de humano otra instancia, incrementando la escalabilidad "Scale Out").
- Como evidencia conclusiva, si navegas por la barra de **Instancias**, atestiguarás la inicialización flamante y sin problemas de la tercera / nueva máquina agregada a las anteriores bajo demanda.

Instances (2/4)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPV6 IPs	Monitoring	Security group name	Key name
WebApp	i-07455c8566fb4bcb9	Running	t3.medium	3/3 checks pass	View alarms +	us-west-2a	ec2-44-251-21-106.us...	44.251.21.106	-	-	disabled	CommandHostSecurity...	vockey
WebApp	i-07455c8566fb4bcb9	Running	t3.micro	3/3 checks pass	View alarms +	us-west-2a	-	-	-	-	disabled	HTTPAccess	-

2 instances selected

Monitoring

Alarm recommendations

Investigate with AI

1h 3h 12h 1d 3d Tw Custom UTC timezone

Explore related

Configure CloudWatch agent

CPU utilization (%)

Network I/O (Bytes)

Network packets (count)

Metadata no token (count)

CPU credit usage (count)

CPU credit balance (count)

☑ **¡Felicidades!** Has concluido este robusto pero metódico laboratorio.